

CONCEPT 22.3

Evolution is supported by an overwhelming amount of scientific evidence (pp. 476–484)

- Researchers have directly observed natural selection leading to evolution in many studies in diverse organisms.
- Organisms share characteristics because of common descent (**homology**) or because natural selection affects independently evolving species in similar environments in similar ways (**convergent evolution**).
- Fossils show that past organisms differed from living organisms, that many species have become extinct, and that species have evolved over long periods of time; fossils also document the evolutionary origin of new groups of organisms.
- Evolutionary theory can explain some **biogeographic** patterns.

? Summarize the different lines of evidence supporting the hypothesis that cetaceans descended from land mammals and are closely related to even-toed ungulates.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the Mastering Biology eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Which of the following is an observation or inference on which natural selection is based?
(A) Individuals do not vary in their heritable characteristics.
(B) Only well-adapted individuals produce offspring.
(C) Species produce more offspring than the environment can support.
(D) Nearly all of each individual's offspring will survive and reproduce.
2. Which of the following observations helped Darwin shape his concept of descent with modification?
(A) Species diversity declines farther from the equator.
(B) Fewer species live on islands than on the nearest continents.
(C) Birds live on islands located farther from the mainland than the birds' maximum nonstop flight distance.
(D) South American temperate plants are more similar to South American tropical plants than to European temperate ones.

Levels 3-4: Applying/Analyzing

3. Within six months of effectively using methicillin to treat *S. aureus* infections in a community, all new *S. aureus* infections were caused by a resistant strain (MRSA). How can this best be explained?
(A) A patient must have become infected with MRSA from another community.
(B) In response to the drug, *S. aureus* began making drug-resistant versions of the protein targeted by the drug.
(C) Some drug-resistant bacteria were present at the start of treatment, and natural selection increased their frequency.
(D) *S. aureus* evolved to resist vaccines.

4. DNA sequences in many human genes are very similar to the sequences of corresponding genes in chimpanzees. The most likely explanation for this result is that
(A) humans and chimpanzees share a relatively recent common ancestor.
(B) humans evolved from chimpanzees.
(C) chimpanzees evolved from humans.
(D) convergent evolution led to the DNA similarities.
5. The upper forelimbs of humans and bats have fairly similar skeletal structures, whereas the corresponding bones in whales have very different shapes and proportions. However, genetic data suggest that all three kinds of organisms diverged from a common ancestor at about the same time. Which of the following is the most likely explanation for these data?
(A) Forelimb evolution was adaptive in people and bats, but not in whales.
(B) Natural selection in an aquatic environment resulted in significant changes to whale forelimb anatomy.
(C) Genes mutate faster in bats than in humans or whales.
(D) Whales are not properly classified as mammals.

Levels 5-6: Evaluating/Creating

6. **EVOLUTION CONNECTION** Explain why anatomical and molecular features often fit a similar nested pattern. In addition, describe a process that can cause this not to be the case.
7. **SCIENTIFIC INQUIRY • DRAW IT** Mosquitoes resistant to the pesticide DDT first appeared in India in 1959, but now are found throughout the world. (a) Graph the data in the table below. (b) After examining the graph, hypothesize why the percentage of mosquitoes resistant to DDT rose rapidly. (c) Suggest an explanation for the global spread of DDT resistance.

Month	0	8	12
Mosquitoes Resistant* to DDT	4%	45%	77%

*Mosquitoes were considered resistant if they were not killed within 1 hour of receiving a dose of 4% DDT.

Data from C. F. Curtis et al., Selection for and against insecticide resistance and possible methods of inhibiting the evolution of resistance in mosquitoes, *Ecological Entomology* 3:273–287 (1978).

8. **WRITE ABOUT A THEME: INTERACTIONS** Write a short essay (about 100–150 words) evaluating whether changes to an organism's physical environment are likely to result in evolutionary change. Use an example to support your reasoning.
9. **SYNTHESIZE YOUR KNOWLEDGE**



This honey pot ant (genus *Myrmecocystus*) can store liquid food inside its expandable abdomen. Consider other ants you are familiar with, and explain how a honey pot ant exemplifies three key features of life: adaptation, unity, and diversity.

For selected answers, see Appendix A.

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How are some coral reefs responding to climate change?

Go to "Take the Heat" at

www.scienceintheclassroom.org.

➔ **Instructors:** Questions can be assigned in Mastering Biology.